Dr. Martin JACQUET

Postdoctoral Researcher

PhD in Robotics
Engineer in Computer Sciences and Applied Mathematics

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Experiences & Education

Sep 2025 – present

Postdoctoral Researcher, at INRIA - RAINBOW team, Rennes, France
Perception-based physical interaction with aerial robots
With Dr. Marco Tognon – ANR-FlyHandyBot project
- Nonlinear control for vision-constrained tasks
- Hybrid/Stochastic NMPC, MPPI
- Integration of onboard lidar-inertial state estimator
- ROS2, GenoM; Python, C++ programming

2020 – present

Temporary Lecturer
- At ENS Rennes (M1, M2):
  - (2026) 28 hours of lectures and practicals – Nonlinear control
  - (2025) 6 hours of lectures – Aerial robotics: modeling and control
- At NTNU (M1):
  - (2023) 10 hours of lectures and practicals – ROS development
- At INP-ENSEEIHT (L3, M1):
  - (2021) 31 hours of seminars and practicals – C programming, Android programming
  - (2020) 21 hours of practicals – Android programming

Feb 2023 – Jun 2025

Postdoctoral Researcher, at NTNU - Autonomous Robots Lab, Trondheim, Norway
Perception-based control methods for collision-free navigation in unknown, complex and cluttered environments
With Prof. Kostas Alexis – EU-Digiforest project
- Deep learning for latent state/environment representations
- Interplaying neural methods with principled control methods (NMPC, CBF) for provable safety
- Software integration and Autonomy Stack development
- Neural world models, Reinforcement Learning
- ROS2, ROS, PX4; python, C++ programming
- Robotics field-testing campaigns: hardware deployment, mission operator in forest environments

Jan 2019 – Oct 2022

PhD in Robotics, at LAAS-CNRS - RIS group, Toulouse, France
Methods for Online Predictive Control of Multi-rotor Aerial Robots with Perception-driven Tasks subject to Sensing and Actuation Constraints
Advised by Prof. Antonio Franchi – ARN-MuRoPhen project
- Perception-constrained NMPC, Multi Aerial Robot Systems
- Active Perception methods, Observation uncertainty estimation
- ROS, GenoM; Python, Matlab and C++ programming
- Involvement in the 2020 MBZIRC robotics competition: perception-related software programming
  →**Bronze medal** in the Grand Challenge competition

Mar 2017 – Jun 2018

Research Engineer at LAAS-CNRS - RAP group, Toulouse, France
- Collaborative project with the French Wine Institute (Gaillac) and Naïo Technology (Toulouse)
- Deep Learning-based detection of trunks and leaf diseases on wine images and videos
- Semantic segmentation, object detection, spatio-temporal classification
- Python and C++ programming, libraries Keras and TensorFlow

Summer 2016

6 months engineering internship at ATOS, Toulouse, France
Detection and tracking of icebergs in satellite images
- Semantic segmentation, classifications, temporal tracking
- C++ programming, web technologies (API development, SQL database, Docker, Kubernetes)

References and additional information available upon request

Summer 2015 — **3 months research internship at IRIT**, Toulouse, France
Search of symmetries for 3D-meshes compression
- Transposition of 2D algorithms in 3D for feature extraction (SIFT)
- Statistical evaluation methodology
- Matlab programming

Summer 2014 — **Work training period in a Japanese hotel, as a cleaning person**
Kamogawa Hotel Mikazuki, Kominato, Chiba, Japan

2013 – 2016 — **Engineering degree, at INP-ENSEEIHT**, Toulouse, France
Computer sciences and applied mathematics department
- Audiovisual data processing - Computer vision - 3D Modeling - Virtual Reality
- Mathematics - Statistics/Probabilities - Linear algebra - Optimization
- Computer sciences - Software programming
- School semester at Polytechnique Montreal, Canada; Courses in multimedia technologies
- 2 months master's project : Building a 3D model of a scene from a flow of RGBD data

2011 – 2013 — **Preparatory school, PSI* (Physics and Engineer Sciences)**
Lycée Champollion, Grenoble, France

2011 — **Scientific Baccalauréat**
Lycée Pierre Termier, Grenoble, France

Scientific Publications

Citation count and data as of 20/10/2026
Total citations: 235
h-index: 8

Journal Publications

preprint, 2026 –Cited: 1 — M. Dharmadhikari, N. Khedekar, M. Kulkarni, M. Nissov, **M. Jacquet**, *et al.*, *The Unified Autonomy Stack: Toward a Blueprint for Generalizable Robot Autonomy*

IJRR, 2025 –Cited: 12 — **M. Jacquet**, M. Harms, K. Alexis, *Neural NMPC through Signed Distance Field Encoding for Collision Avoidance*

RAL, 2022 –Cited: 22 — **M. Jacquet**, M. Kivits, H. Das, A. Franchi, *Motor-level N-MPC for Cooperative Active Perception with Multiple Heterogeneous UAVs*

RAL, 2021 –Cited: 44 — **M. Jacquet**, A. Franchi, *Motor and Perception Constrained NMPC for Torque-Controlled Generic Aerial Vehicles*
– **Finalist for Best Paper Award** in Unmanned Aerial Vehicles at ICRA'21

RAL is the IEEE reference for timely publications of major results in Robotics
Impact Factor: 5.3 – Journal Citation Indicator: 1.22
IJRR is among the two major venues for comprehensive Robotics publications
Impact Factor: 5.0 – Journal Citation Indicator: 1.36

Conference Publications

ICRA'25 –Cited: 19 — M. Harms, **M. Jacquet**, K. Alexis, *Safe Quadrotor Navigation using Composite Control Barrier Functions*

ICUAS'25 –Cited: 2 — N. Misyats, M. Harms, M. Nissov, **M. Jacquet**, K. Alexis, *Embedded Safe Reactive Navigation for Multirotors Systems using Control Barrier Functions*

IROS'24 –Cited: 38 — M. Harms, M. Kulkarni, N. Khedekar, **M. Jacquet**, K. Alexis, *Neural Control Barrier Functions for Safe Navigation*

ICRA'24 –Cited: 20 — **M. Jacquet**, K. Alexis, *N-MPC for Deep Neural Network-Based Collision Avoidance exploiting Depth Images*

IROS'22 –Cited: 26 — G. Corsini, **M. Jacquet**, H. Das, A. Afifi, D. Sidobre, A. Franchi, *Nonlinear Model Predictive Control for Human-Robot Handover with Application to the Aerial Case*

IROS'22 –Cited: 6 — **M. Jacquet**, A. Franchi, *Enforcing Vision-Based Localization using Perception Constrained N-MPC for Multi-Rotor Aerial Vehicles*

AIRPHARO'21 –Cited: 5 — G. Corsini, **M. Jacquet**, AE. Jimenez-Cano, A. Afifi, D. Sidobre, A. Franchi, *A General Control Architecture for Visual Servoing and Physical Interaction Tasks for Fully-actuated Aerial Vehicles*

References and additional information available upon request

ICRA'20 –Cited: 28 — **M. Jacquet**, G. Corsini, D. Bicego, A. Franchi, *Perception-constrained and Motor-level Nonlinear MPC for both Underactuated and Tilted-propeller UAVS*

ICRA and IROS are the two IEEE flagship conferences in Robotics

Book Chapters

Springer, 2026 — M. Camurri, E. Tomelleri, M. Mattamala, S. Barbas Laina, **M. Jacquet**, *et al.*, *DigiForest: Digital Analytics and Robotics for Sustainable Forestry*, in *Recent Advances in Robotic Perception for Forestry*

Editorial services

Reviewer — Journals: Nature Communications, T-RO, IJRR, RAL
Conferences: RSS, ICRA, IROS, ICUAS, ECC

Associate Editor — IROS (2026)

Program Committee — ECMR (2027)

Co-supervision of Master's Students

Sep 2024 – Dec 2024 — Nazar Misyats, ENS Rennes, France
Development of a Safe Embedded Drone Controller based on Control Barrier Functions
Role: Topic definition, theoretical supervision, experimental hardware validation

Jan 2024 – Jul 2024 — Ask Espensønn Øren, NTNU, Norway
Multi-modal Learning-based Navigation
Role: Topic definition, technical & theoretical supervision

Jan 2024 – Jul 2024 — Mathias Skogli, NTNU, Norway
Reinforcement Learning for Autonomous Safe Navigation in Dynamic and Cluttered Environments Exploiting Dynamics-Aware Embeddings of Depth Images
Role: Topic definition, technical & theoretical supervision

Jan 2020 – Jul 2021 — Max Kivits, University of Twente, The Netherlands
Heterogeneous Cooperative Target Tracking using Nonlinear Model Predictive Control
Role: Topic definition, technical supervision, experimental hardware validation

Jan 2020 – Jul 2020 — Vamshi Kodipaka, University of Bourgogne, France
Vision based Multi-robot Tracking of Target with Aerial and Ground Robots
Role: Topic definition, technical supervision

Maintained Open Source Software

Autonomy Stack — Comprehensive Unified Autonomy Stack for Robots – Dev., auth. of related paper – 241 stars

SDF-NMPC — Neural MPC for collision avoidance – Main dev., first auth. of related paper – 55 stars

CCBF — Composite CBF for collision avoidance – Main dev., auth. of related paper – 14 stars

PX4-CBF — Embedded impl. of CCBF in PX4 Autopilot – Dev., auth. of related paper – 13 stars

All my papers also have an open-source implementation for reproducibility (not actively maintained)

Technical skills

Scientific — Neural Networks, Implicit neural representations
Nonlinear optimal control
Pose estimation, uncertainties, Kalman filtering
Deep learning for image and video processing
Mathematical modeling, physical parameters estimation

Languages — Python, C++, C, Matlab

Programming — ROS2, ROS, GenoM, PX4, LateX, Git, Docker, Docker-compose

Librairies — PyTorch, OpenCV, Eigen

Hardware — Camera (calibration, data acquisition), Lidars, IMUs, Custom drones & electronics

Languages

French	Native spealer
English	C2, fluent written and oral communication; Toeic obtained in 2015 (940) Professional daily communication at work (working in international teams since 2019)
Spanish	High-school level (B1)
Japanese	Advanced Beginner
Norwegian	Beginner
Italian	Beginner

Various info

Hobbies	Hobby software programming, cycling, brewing, tabletop role-playing, video games, cinema, photography, science-fiction